

Ideals and Varieties

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This is my notes on the textbook *Ideals, Varieties, and Algorithms* [1]. Algebraic geometry is a fundamental tool for polynomial optimization: the equality constraints in an optimization problem define varieties. Understanding these constraint sets algebraically is essential for solving constrained polynomial optimization problems.

1 Algebra and Geometry

1.1 Affine Varieties

A *monomial* in $x_1, \dots, x_n$ is a product of the form $x_1^{\alpha_1} \cdot x_2^{\alpha_2} \cdots x_n^{\alpha_n}$, where $\alpha_i \in \mathbb{Z}_+$.

- $\alpha \triangleq (\alpha_1, \dots, \alpha_n) \in \mathbb{Z}_+^n$,
- $x^\alpha \triangleq x_1^{\alpha_1} \cdots x_n^{\alpha_n}$,
- $|\alpha| = \sum_{i=1}^n \alpha_i$ is the *degree* of the monomial.

A *polynomial* f in $x_1, \dots, x_n$ with coefficients in a field $\mathbb{F}$ is a finite linear combination of monomials in the form

$$f = \sum_{\alpha} c_{\alpha} x^{\alpha}, \quad c_{\alpha} \in \mathbb{F},$$

where the sum is over a finite number of n -tuples $\alpha = (\alpha_1, \dots, \alpha_n)$.

- $\deg(f)$ is the maximum $|\alpha|$ such that $c_{\alpha} \neq 0$,
- $\mathbb{F}[x_1, \dots, x_n]$ is the set of all polynomials in $x_1, \dots, x_n$ with coefficients in $\mathbb{F}$.

Given a field $\mathbb{F}$ and a positive integer n , the n -dimensional *affine space* over $\mathbb{F}$ is the set

$$\mathbb{F}^n = \{(a_1, \dots, a_n) \mid a_1, \dots, a_n \in \mathbb{F}\}.$$

Note that the difference between linear and affine space is whether it contains 0 or not. We can now define the geometric object that will be studied in this note:

Definition 1.1: Affine Variety

Let $\mathbb{F}$ be a field, and let $f_1, \dots, f_s$ be polynomials in $\mathbb{F}[x_1, \dots, x_n]$. Then the affine variety defined by $f_1, \dots, f_s$ is

$$V(f_1, \dots, f_s) = \{(a_1, \dots, a_n) \in \mathbb{F}^n \mid f_i(a_1, \dots, a_n) = 0, i \in [s]\}.$$

$V(f_1, \dots, f_s) \subseteq \mathbb{F}^n$ is the set of solutions of the polynomial equation system

$$f_1(x_1, \dots, x_n) = \cdots = f_s(x_1, \dots, x_n) = 0. \quad (1.1)$$

For example, the affine variety $V(x_1^2 + x_2^2 - 1)$ is the unit circle in $\mathbb{R}^2$.

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1.2 Parametrizations of Affine Varieties

We can describe an affine variety by the solutions of a polynomial system. For example, the unit circle $x_1^2 + x_2^2 = 1$ has different parametrizations:

- $x_1 = \cos(t), x_2 = \sin(t),$
- $x_1 = \frac{1-t^2}{1+t^2}, x_2 = \frac{2t}{1+t^2}.$

Note that the second parametrization does not cover the point $(-1, 0)$. Furthermore, it involves rational functions.

Definition 1.2: Rational function

Let $\mathbb{F}$ be a field. A rational function in $t_1, \dots, t_m$ with coefficients in $\mathbb{F}$ is a quotient f/g of two polynomials $f, g \in \mathbb{F}[t_1, \dots, t_m]$, where $g \neq 0$.

- Two rational functions f/g and f'/g' are equal if $fg' = f'g$.
- $\mathbb{F}(t_1, \dots, t_m)$ is the set of all rational functions in $t_1, \dots, t_m$ with coefficients in $\mathbb{F}$.

Given a variety $V(f_1, \dots, f_s) \subseteq \mathbb{F}^n$, a *rational parametric representation* of V consists of rational functions $r_1, \dots, r_n \in \mathbb{F}(t_1, \dots, t_m)$ such that the points

$$\begin{aligned} x_1 &= r_1(t_1, \dots, t_m) \\ &\vdots \\ x_n &= r_n(t_1, \dots, t_m) \end{aligned} \tag{1.2}$$

lie in V . We call the special case where $r_1, r_2, \dots, r_n$ are just polynomials a *polynomial parametric representation*. To this end, we have two ways of describing a variety:

- *implicit representation*: the polynomial equation systems (1.1),
- *parametric representation*: the rational parametric representation (1.2).

This leads to two questions: Does every variety have a parametric representation, and given a parametric representation, can we find its implicit representation? The answer to the first question is no, in fact, most varieties do not have a parametric representation. The answer to the second question is yes, and will be discussed in Section 2.

1.3 Ideals

Definition 1.3: Ideal

A subset $I \subseteq \mathbb{F}[x_1, \dots, x_n]$ is an ideal if it satisfies:

- (i) $0 \in I$.
- (ii) $f, g \in I \implies f + g \in I$.
- (iii) $f \in I, h \in \mathbb{F}[x_1, \dots, x_n] \implies hf \in I$.

A natural example is the ideal generated by finite polynomials. Let $f_1, \dots, f_s \in \mathbb{F}[x_1, \dots, x_n]$, the ideal generated by $f_1, \dots, f_s$ is denoted by

$$\langle f_1, \dots, f_s \rangle = \left\{ \sum_{i=1}^s h_i f_i \mid h_1, \dots, h_s \in \mathbb{F}[x_1, \dots, x_n] \right\}.$$

A nice property of an ideal generated by polynomials is that every equation $p = 0$ where $p \in \langle f_1, \dots, f_s \rangle$ is a polynomial combination¹ of the polynomial equation system $f_1 = \dots = f_s = 0$. We say that an ideal I is finitely generated if there exists $f_1, \dots, f_s \in \mathbb{F}[x_1, \dots, x_n]$ such that $I = \langle f_1, \dots, f_s \rangle$, and we say that $f_1, \dots, f_s$ are a basis of I .

Proposition 1.4

If $f_1, \dots, f_s$ and $g_1, \dots, g_t$ are bases of the same ideal in $\mathbb{F}[x_1, \dots, x_n]$ such that $\langle f_1, \dots, f_s \rangle = \langle g_1, \dots, g_t \rangle$, then $V(f_1, \dots, f_s) = V(g_1, \dots, g_t)$.

The fact that changing the basis does not affect a variety shows that affine varieties are determined by ideals, not polynomials. That is, given a variety $V(f_1, \dots, f_s) \subseteq \mathbb{F}^n$, we know that $f_1, \dots, f_s$ vanish on V , but they are not the only ones. We next consider the set of all polynomials that vanish on a given variety. Let $V \subseteq \mathbb{F}^n$ be an affine variety, the ideal of V is denoted by

$$I(V) = \{f \in \mathbb{F}[x_1, \dots, x_n] \mid f(a_1, \dots, a_n) = 0, (a_1, \dots, a_n) \in V\}.$$

It is trivial that $\langle f_1, \dots, f_s \rangle \subseteq I(V(f_1, \dots, f_s))$ since polynomials that vanish on V is not necessary a polynomial combination of $f_1, \dots, f_s$. For example, consider the ideal $\langle x_1^2, x_2^2 \rangle$, we have the variety $V(x_1^2, x_2^2) = \{(0, 0)\}$. One can easily verify that $I(\{(0, 0)\}) = \langle x_1, x_2 \rangle$, and since $\langle x_1^2, x_2^2 \rangle$ does not contain degree 1 polynomials, we have $\langle x_1^2, x_2^2 \rangle \subset \langle x_1, x_2 \rangle = I(V(x_1^2, x_2^2))$.

2 Gröbner Basis

In this section, we will focus on the following problems:

1. Does every ideal $I \subseteq \mathbb{F}[x_1, \dots, x_n]$ has a finite basis, i.e., can every ideal be written as $\langle f_1, \dots, f_s \rangle$ for some $f_1, \dots, f_s \in \mathbb{F}[x_1, \dots, x_n]$?
2. Given $f \in \mathbb{F}[x_1, \dots, x_n]$, can we determine if $f \in \langle f_1, \dots, f_s \rangle$ for some $f_1, \dots, f_s \in \mathbb{F}[x_1, \dots, x_n]$?
3. How to solve the polynomial equation system $f_1(x_1, \dots, x_n) = \dots = f_s(x_1, \dots, x_n) = 0$?
4. Given an affine variety $V \subseteq \mathbb{F}^n$ with parametric representation, how to find its implicit representation?

2.1 Orderings of Monomials

Before we get into the problems, we need to first decide the ordering of terms in polynomials. Recall that monomials $x^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n} \in \mathbb{F}[x_1, \dots, x_n]$ has a one-to-one correspondence with tuples $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{Z}_+^n$, any ordering on $\mathbb{Z}_+^n$ will give us an ordering on monomials. A *monomial ordering* $>$ on $\mathbb{F}[x_1, \dots, x_n]$ is a relation $>$ on $\mathbb{Z}_+^n$, or equivalently, a relation on the set of monomials x^α , satisfying: (i) $>$ is a linear/total ordering, (ii) $\alpha > \beta, \gamma \in \mathbb{Z}_+^n \implies \alpha + \gamma > \beta + \gamma$, and (iii) $>$ is a well-ordering, i.e., if $A \subseteq \mathbb{Z}_+^n$ is nonempty, then there exists $\alpha \in A$ such that $\beta > \alpha$ for every $\beta \neq \alpha$ where $\beta \in A$. We introduce two examples:

- *Lexicographic Order*: Let $\alpha = (\alpha_1, \dots, \alpha_n)$ and $\beta = (\beta_1, \dots, \beta_n)$ be in $\mathbb{Z}_+^n$. We say $\alpha >_{\text{lex}} \beta$ if the leftmost nonzero entry of $\alpha - \beta$ is positive.
- *Graded Lexicographic Order*: Let $\alpha, \beta \in \mathbb{Z}_+^n$. We say $\alpha >_{\text{grlex}} \beta$ if either $|\alpha| > |\beta|$ or $|\alpha| = |\beta|$ and $\alpha >_{\text{lex}} \beta$.

¹Ideals in $\mathbb{F}[x_1, \dots, x_n]$ are analogous to subspaces in vector space.

The lex order does not take the total degree into account. For instance, given $f = 4x_1x_2^2x_3 + 4x_3^2 - 5x_1^3 + 7x_1^2x_3^2$, we write f in decreasing order with respect to:

- lex order: $f = -5x_1^3 + 7x_1^2x_3^2 + 4x_1x_2^2x_3 + 4x_3^2$,
- grlex order: $f = 7x_1^2x_3^2 + 4x_1x_2^2x_3 - 5x_1^3 + 4x_3^2$.

2.2 Division Algorithm

We first define some notation. Let $f = \sum_{\alpha} c_{\alpha} x^{\alpha}$ be a nonzero polynomial in $\mathbb{F}[x_1, \dots, x_n]$ and let $>$ be a monomial order.

- The multidegree of f is $\text{multideg}(f) = \max(\alpha \in \mathbb{Z}_+^n \mid c_{\alpha} \neq 0)$.
- The leading coefficient of f is $\text{LC}(f) = c_{\text{multideg}(f)}$.
- The leading monomial of f is $\text{LM}(f) = x^{\text{multideg}(f)}$.
- The leading term of f is $\text{LT}(f) = \text{LC}(f) \times \text{LM}(f)$.

Lemma 2.1

Let $f, g \in \mathbb{F}[x_1, \dots, x_n]$ be nonzero polynomials. Then:

- (i) $\text{multideg}(fg) = \text{multideg}(f) + \text{multideg}(g)$.
- (ii) If $f + g \neq 0$, then $\text{multideg}(f + g) \leq \max(\text{multideg}(f), \text{multideg}(g))$. In addition, if $\text{multideg}(f) \neq \text{multideg}(g)$, then equality holds.

Algorithm 1 Division Algorithm

Input: $f, f_1, \dots, f_s$

Output: $q_1, \dots, q_s, r$

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1: initialize  $q_1 = \dots = q_s = 0$ ,  $r = 0$ , and  $p = f$ 
2: while  $p \neq 0$  do
3:   set  $i = 1$  and  $\text{div\_flag} = \text{false}$ 
4:   while  $i \leq s$  and  $\text{div\_flag} = \text{false}$  do
5:     if if  $\text{LT}(f_i)$  divides  $\text{LT}(p)$  then
6:       update  $q_i \leftarrow q_i + \frac{\text{LT}(p)}{\text{LT}(f_i)}$  and  $p \leftarrow p - \frac{\text{LT}(p)}{\text{LT}(f_i)} f_i$ 
7:       set  $\text{div\_flag} = \text{true}$ 
8:     else
9:        $i \leftarrow i + 1$ 
10:    end if
11:  end while
12:  if  $\text{div\_flag} = \text{false}$  then
13:    update  $r \leftarrow r + \text{LT}(p)$  and  $p \leftarrow p - \text{LT}(p)$ 
14:  end if
15: end while
16: return  $q_1, \dots, q_s, r$ 

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The goal is to divide $f \in \mathbb{F}[x_1, \dots, x_n]$ by $f_1, \dots, f_s \in \mathbb{F}[x_1, \dots, x_n]$, i.e., express f as

$$f = q_1 f_1 + \dots + q_s f_s + r,$$

where the quotients $q_1, \dots, q_s$ and remainder r lie in $\mathbb{F}[x_1, \dots, x_n]$. The full procedure is depicted in Algorithm 1, followed by two examples².

²We sometimes use $x \triangleq x_1, y \triangleq x_2, z \triangleq x_3, \dots$ in examples for notation simplicity

Example 2.2

Divide $f = x^2y + xy^2 + y^2$ by $f_1 = xy - 1$ and $f_2 = y^2 - 1$ with lex order.

$$\begin{array}{rcl}
 q_1 & x + y & r \\
 q_2 & 1 & \\
 xy - 1 & \sqrt{x^2y + xy^2 + y^2} & \\
 y^2 - 1 & x^2y - x & \\
 \hline
 & xy^2 + x + y^2 & \\
 & xy^2 - y & \\
 \hline
 & x + y^2 + y & \rightarrow x \\
 & y^2 + y & \\
 & y^2 - 1 & \\
 \hline
 & y + 1 & \rightarrow x + y + 1
 \end{array}$$

We obtain $f = q_1f_1 + q_2f_2 + r$, where $q_1 = x + y$, $q_2 = 1$, and $r = x + y + 1$. Now we rearrange the divisors, i.e., divide f by $f_1 = y^2 - 1$ and $f_2 = xy - 1$.

$$\begin{array}{rcl}
 q_1 & x + 1 & r \\
 q_2 & x & \\
 y^2 - 1 & \sqrt{x^2y + xy^2 + y^2} & \\
 xy - 1 & x^2y - x & \\
 \hline
 & xy^2 + x + y^2 & \\
 & xy^2 - x & \\
 \hline
 & 2x + y^2 & \rightarrow 2x \\
 & y^2 & \\
 & y^2 - 1 & \\
 \hline
 & 1 & \rightarrow 2x + 1
 \end{array}$$

We obtain $f = q_1f_1 + q_2f_2 + r$, where $q_1 = x + 1$, $q_2 = x$, and $r = 2x + 1$.

Example 2.2 shows that the remainder is not unique, and can be changed if we rearrange the divisors. Note that if $r = 0$, then $f = q_1f_1 + \dots + q_sf_s \in \langle f_1, \dots, f_s \rangle$. Therefore, $r = 0$ is a sufficient condition for f being in an ideal. Intuitively, instead of working with $f_1, \dots, f_s$, we consider the ideal $I = \langle f_1, \dots, f_s \rangle$. This allows us to go from $f_1, \dots, f_s$ to another basis of I . We will later show that Gröbner basis has a nice property about the uniqueness of remainder.

2.3 Monomial Ideals

Definition 2.3

An ideal $I \subseteq \mathbb{F}[x_1, \dots, x_n]$ is a monomial ideal if there exists a subset $A \subseteq \mathbb{Z}_+^n$ such that I consists of all polynomials which are finite sums of the form $\sum_{\alpha \in A} h_\alpha x^\alpha$, where $h_\alpha \in \mathbb{F}[x_1, \dots, x_n]$. In this case, we write $I = \langle x^\alpha \mid \alpha \in A \rangle$.

Let $I = \langle x^\alpha \mid \alpha \in A \rangle$ be a monomial ideal, then one can easily show that a monomial $x^\beta \in I$ if and only if x^β is divisible by x^α for some $\alpha \in A$.

Theorem 2.4: Dickson's Lemma

Let $I = \langle x^\alpha \mid \alpha \in A \rangle \in \mathbb{F}[x_1, \dots, x_n]$ be a monomial ideal. Then I can be written in the form $I = \langle x^{\alpha(1)}, \dots, x^{\alpha(s)} \rangle$, where $\alpha(1), \dots, \alpha(s) \in A$, i.e., I has a finite basis.

Dickson's Lemma solves Problem 1 for monomial ideals. Furthermore, we also solved Problem 2 for monomial ideals. Let $I = \langle x^{\alpha(1)}, \dots, x^{\alpha(s)} \rangle$ be a monomial ideal. Since $r = 0$ is a sufficient condition for $f \in I$, we have $r = 0 \implies f \in I$. On the other hand, let $f \in I$, since each term is divisible by one of the generators, the division algorithm will always return $r = 0$. That is, $f \in I \implies r = 0$. We can conclude that $f \in I$ if and only if the remainder of f division by $x^{\alpha(1)}, \dots, x^{\alpha(s)}$ is zero.

Proposition 2.5

A monomial ideal $I \in \mathbb{F}[x_1, \dots, x_n]$ has a unique basis such that $x^{\alpha(i)}$ does not divide $x^{\alpha(j)}$ for $i \neq j$. This basis is called the *minimal basis* of I .

2.4 The Hilbert Basis Theorem and Gröbner basis

We have shown the solution to Problem 1 for monomial ideals in Section 2.3, we next extend it for general ideals. Since the leading terms play an important role in the division algorithm, a natural idea is to consider only the leading terms, for example, the monomial ideal generated by the leading terms $\langle \text{LT}(f_1), \dots, \text{LT}(f_s) \rangle$. We next define another ideal, the *ideal of leading terms*. Let $I \subseteq \mathbb{F}[x_1, \dots, x_n]$ be an ideal other than $\{0\}$, we denote by $\text{LT}(I)$ the set of leading terms of nonzero elements of I . Thus,

$$\text{LT}(I) = \{cx^\alpha \mid \exists f \in I \setminus \{0\} \text{ s.t. } \text{LT}(f) = cx^\alpha\}.$$

We denote by $\langle \text{LT}(I) \rangle$ the ideal generated by the elements of $\text{LT}(I)$. Note that $\langle \text{LT}(I) \rangle$ is also a monomial ideal. Let $I = \langle f_1, \dots, f_s \rangle$, it follows that $\langle \text{LT}(f_1), \dots, \text{LT}(f_s) \rangle \subseteq \langle \text{LT}(I) \rangle$ since the formal ideal is generated by the leading terms of $f_1, \dots, f_s$, while the latter ideal is generated by all leading terms of $f \in I$.

Example 2.6

Let $I = \langle f_1, f_2 \rangle$, where $f_1 = x^3 - 2xy$ and $f_2 = x^2y - 2y^2 + x$ with the grlex ordering. We have

$$-y \underbrace{(x^3 - 2xy)}_{f_1} + x \underbrace{(x^2y - 2y^2 + x)}_{f_2} = x^2 \in I.$$

Here, $x^2 \in \langle \text{LT}(I) \rangle$ but $x^2 \notin \langle \text{LT}(f_1), \text{LT}(f_2) \rangle$ because the leading terms are *cancelled out*.

Proposition 2.7

Let $I \subseteq \mathbb{F}[x_1, \dots, x_n]$ be an ideal other than $\{0\}$. Then there exists $g_1, \dots, g_t \in I$ such that $\langle \text{LT}(I) \rangle = \langle \text{LT}(g_1), \dots, \text{LT}(g_t) \rangle$.

Proof. Since $\langle \text{LT}(I) \rangle$ is generated by $\text{LM}(g)$ for $g \in I \setminus \{0\}$, by Dickson's Lemma, $\langle \text{LT}(I) \rangle = \langle \text{LM}(g_1), \dots, \text{LM}(g_t) \rangle = \langle \text{LT}(g_1), \dots, \text{LT}(g_t) \rangle$ for some $g_1, \dots, g_t \in I$. $\square$

By Proposition 2.7 and the division algorithm 1, we can show that every ideal has a finite generating set.

Theorem 2.8: Hilbert Basis Theorem

Every ideal $I \subseteq \mathbb{F}[x_1, \dots, x_n]$ has a finite generating set. In other words, $I = \langle g_1, \dots, g_t \rangle$ for some $g_1, \dots, g_t \in I$.

Proof. If $I = \{0\}$, then the finite generating set is $\{0\}$. Let $I \neq \{0\}$, by Proposition 2.7, there exists $g_1, \dots, g_t \in I$ such that $\langle \text{LT}(I) \rangle = \langle \text{LT}(g_1), \dots, \text{LT}(g_t) \rangle$.

- Since $g_1, \dots, g_t \in I$, we have $\langle g_1, \dots, g_t \rangle \subseteq I$.
- Let $f \in I$ and divide f by $g_1, \dots, g_t$, we have $f = q_1g_1 + \dots + q_tg_t + r$. If $r \neq 0$, we have $r = f - q_1g_1 - \dots - q_tg_t \in I$. Furthermore, $\text{LT}(r) \in \langle \text{LT}(I) \rangle = \langle \text{LT}(g_1), \dots, \text{LT}(g_t) \rangle$. Since $\langle \text{LT}(g_1), \dots, \text{LT}(g_t) \rangle$ is a monomial ideal, thus $\text{LT}(r)$ must be divisible by some $\text{LT}(g_i)$. This contradicts the definition of remainder, we conclude that $r = 0$. Therefore, $f = q_1g_1 + \dots + q_tg_t \in \langle g_1, \dots, g_t \rangle$.

Altogether, we have $I = \langle g_1, \dots, g_t \rangle$. □

The Hilbert Basis Theorem solves Problem 1: every ideal $I \subseteq \mathbb{F}[x_1, \dots, x_n]$ can be written as $\langle f_1, \dots, f_s \rangle$ for some $f_1, \dots, f_s \in \mathbb{F}[x_1, \dots, x_n]$. The special basis $\langle g_1, \dots, g_t \rangle$ that is used in the proof (by Dickson's Lemma) is also known as the Gröbner basis.

Definition 2.9: Gröbner Basis

Fix a monomial order on the polynomial ring $\mathbb{F}[x_1, \dots, x_n]$. A finite subset $G = \{g_1, \dots, g_t\}$ of an ideal $I \subseteq \mathbb{F}[x_1, \dots, x_n]$ other than $\{0\}$ is said to be a Gröbner basis if

$$\langle \text{LT}(g_1), \dots, \text{LT}(g_t) \rangle = \langle \text{LT}(I) \rangle.$$

We define the empty set $\emptyset$ to be the Gröbner basis of $\{0\}$.

It follows by the Hilbert Basis Theorem that every ideal has a Gröbner basis. To this end, we have considered affine varieties as the sets of solutions of system of finite equations:

$$V(f_1, \dots, f_s) = \{(a_1, \dots, a_n) \in \mathbb{F}^n \mid f_i(a_1, \dots, a_n) = 0, i \in [s]\}.$$

The Hilbert Basis Theorem shows that it also makes sense to define affine variety by an ideal

$$V(I) = \{(a_1, \dots, a_n) \in \mathbb{F}^n \mid f(a_1, \dots, a_n) = 0, f \in I\}.$$

A nonzero ideal contains infinitely many different polynomials, but $V(I)$ can be defined by a finite set of equations. Note that if $I = \langle f_1, \dots, f_s \rangle$, then $V(I) = V(f_1, \dots, f_s)$. We next show that Gröbner bases have a nice property on the uniqueness of remainder, which can be used to determine whether a given basis is a Gröbner basis or not.

2.5 Buchberger's Criterion

Proposition 2.10

Let $I \subseteq \mathbb{F}[x_1, \dots, x_n]$ be an ideal and $G = \{g_1, \dots, g_t\}$ be a Gröbner basis for I . Then given $f \in \mathbb{F}[x_1, \dots, x_n]$, there exists a *unique* remainder $r \in \mathbb{F}[x_1, \dots, x_n]$ on division of f by G .

Proof. Suppose $f = g + r = g' + r'$, where $g, g' \in I$. Since $r - r' = g' - g \in I$ and G is a Gröbner basis, if $r \neq r'$, then $\text{LT}(r - r') \in \langle \text{LT}(I) \rangle = \langle \text{LT}(g_1), \dots, \text{LT}(g_t) \rangle$. It follows that $\text{LT}(r - r')$ is divisible by some $\text{LT}(g_i)$, which contradicts because no term of r and r' is divisible by $\text{LT}(g_i)$. Therefore, $r = r'$. □

As a corollary of Proposition 2.10, we immediately have the following criterion to determine whether a given polynomial lies in an ideal or not.

Corollary 2.11

Let $G = \{g_1, \dots, g_t\}$ be a Gröbner basis for an ideal $I \subseteq \mathbb{F}[x_1, \dots, x_n]$ and let $f \in \mathbb{F}[x_1, \dots, x_n]$. Then $f \in I$ if and only if the remainder of the division of f by G is zero.

Let r be the remainder, recall that $r = 0$ is a sufficient condition for $f \in I$. Since the remainder is unique (using Gröbner basis), $r = 0$ becomes sufficient and necessary for $f \in I$. Therefore, given a Gröbner basis, we can now solve Problem 2 by checking the remainder. We will introduce how to find a Gröbner basis in Section 2.6. Before that, we first discuss how to check if a given basis is a Gröbner basis.

Definition 2.12: S-Polynomial

Let $f, g \in \mathbb{F}[x_1, \dots, x_n]$ be nonzero polynomials.

- (i) Suppose that $\text{multideg}(f) = \alpha$ and $\text{multideg}(g) = \beta$, let $\gamma = (\gamma_1, \dots, \gamma_n)$, where $\gamma_i = \max(\alpha_i, \beta_i)$ for each i . We call $x^\gamma = \text{lcm}(\text{LM}(f), \text{LM}(g))$ the *least common multiple* of $\text{LM}(f)$ and $\text{LM}(g)$.
- (ii) The S-polynomial of f and g is the combination

$$S(f, g) = \frac{x^\gamma}{\text{LT}(f)}f - \frac{x^\gamma}{\text{LT}(g)}g.$$

S-polynomial cancels out the leading terms. For example, let $f = x_1^3x_2 - x_1^2x_2^3 + x_1$ and $g = 3x_1^4 + x_2^2$. Then $\gamma = (4, 2)$ and

$$\begin{aligned} S(f, g) &= \frac{x_1^4x_2^2}{x_1^3x_2^2}(x_1^3x_2 - x_1^2x_2^3 + x_1) - \frac{x_1^4x_2^2}{3x_1^4}(3x_1^4 + x_2^2) \\ &= x_1(x_1^3x_2 - x_1^2x_2^3 + x_1) - \frac{x_2}{3}(3x_1^4 + x_2^2) \\ &= (x_1^4x_2 - x_1^3x_2^3 + x_1^2) - (x_1^4x_2 - \frac{x_2^3}{3}) \\ &= -x_1^3x_2^3 + x_1^2 + \frac{x_2^3}{3} \end{aligned}$$

Lemma 2.13

Suppose we have a sum $\sum_{i=1}^s p_i$, where $\text{multideg}(p_i) = \delta \in \mathbb{Z}_+^n$ for all i . If cancellation occurs, i.e., $\text{multideg}(\sum_{i=1}^s p_i) < \delta$, then $\sum_{i=1}^s p_i$ is a linear combination of the S-polynomials $S(p_j, p_l)$ for $1 \leq j, l \leq s$. Furthermore, each $S(p_j, p_l)$ has multidegree less than δ .

Proof. Let $c_i = \text{LC}(p_i)$, since $\text{multideg}(\sum_{i=1}^s p_i) < \delta$, we have $\sum_{i=1}^s c_i = 0$. Since p_i has the same leading monomial for all i , the S polynomial of p_j and p_l reduces to

$$S(p_j, p_l) = \frac{x^\delta}{c_j x^\delta} p_j - \frac{x^\delta}{c_l x^\delta} p_l = \frac{1}{c_j} p_j - \frac{1}{c_l} p_l.$$

Note that $\sum_{i=1}^s c_i = 0$ implies $c_1 + \cdots + c_{s-1} = -c_s$, it follows that

$$\begin{aligned} \sum_{i=1}^{s-1} c_i S(p_i, p_s) &= c_1 \left(\frac{1}{c_1} p_1 - \frac{1}{c_s} p_s \right) + \cdots + \left(\frac{1}{c_{s-1}} p_{s-1} - \frac{1}{c_s} p_s \right) \\ &= p_1 + \cdots + p_{s-1} - \frac{1}{c_s} \underbrace{(c_1 + \cdots + c_{s-1})}_{-c_s} p_s \\ &= p_1 + \cdots + p_{s-1} + p_s \end{aligned}$$

Thus $\sum_{i=1}^s p_i$ is a linear combination of the S-polynomials $S(p_j, p_l)$ for $1 \leq j, l \leq s$. $\square$

When we are adding polynomials, cancellations of leading terms can potentially introduce new/hidden leading terms. S-polynomials adjust this issue by forcing cancellation. Lemma 2.13 shows that all polynomials that cancellation occurs are linear combinations of the S-polynomials so that we don't need to worry about random cancellations. Using the S-polynomials, we can prove the Buchberger's Criterion of checking if a given basis is a Gröbner basis.

Theorem 2.14: Buchberger's Criterion

Let I be a polynomial ideal. Then a basis $G = \{g_1, \dots, g_t\}$ of I is a Gröbner basis if and only if for all pairs $i \neq j$, the remainder of the division of $S(g_i, g_j)$ by G is zero.

Proof. Let G be a Gröbner basis, since $S(g_i, g_j) \in I$, by Corollary 2.11, the remainder of the division of $S(g_i, g_j)$ by G is zero. Conversely, Let $f \in I$ be nonzero, since G is a basis, write $f = \sum_{i=1}^t h_i g_i$. By Lemma 2.13, it follows that

$$\text{multideg}(f) \leq \max(\text{multideg}(h_i g_i) \mid h_i g_i \neq 0).$$

We pick for which $\delta = \max(\text{multideg}(h_i g_i) \mid h_i g_i \neq 0)$ is minimal, i.e., the minimum possible one, it follows that $\text{multideg}(f) \leq \delta$.

- If $\text{multideg}(f) = \delta$, then $\text{multideg}(f) = \text{multideg}(h_i g_i)$ for some i , i.e., $\text{LT}(f)$ is divisible by $\text{LT}(g_i)$. Then $\text{LT}(f) \in \langle \text{LT}(g_1), \dots, \text{LT}(g_t) \rangle$ for any $f \in I \setminus \{0\}$, which implies that

$$\langle \text{LT}(I) \rangle \subseteq \langle \text{LT}(g_1), \dots, \text{LT}(g_t) \rangle.$$

On the other hand, it is trivial that

$$\langle \text{LT}(g_1), \dots, \text{LT}(g_t) \rangle \subseteq \langle \text{LT}(I) \rangle.$$

Altogether, $\langle \text{LT}(g_1), \dots, \text{LT}(g_t) \rangle = \langle \text{LT}(I) \rangle$, thus G is a Gröbner basis.

- If $\text{multideg}(f) < \delta$, we first write

$$\begin{aligned} f &= \sum_{\text{multideg}(h_i g_i) = \delta} h_i g_i + \sum_{\text{multideg}(h_i g_i) < \delta} h_i g_i \\ &= \sum_{\text{multideg}(h_i g_i) = \delta} \text{LT}(h_i) g_i + \sum_{\text{multideg}(h_i g_i) < \delta} (h_i - \text{LT}(h_i)) g_i + \sum_{\text{multideg}(h_i g_i) < \delta} h_i g_i. \end{aligned} \quad (2.1)$$

The second and third term of (2.1) must have multidegree less than δ by construction, and since $\text{multideg}(f) < \delta$, the first term also has multidegree less than δ . Let $p_i \triangleq \text{LT}(h_i) g_i$,

by Lemma 2.13, the first term is a linear combination of S-polynomials $S(p_i, p_j)$. Note that

$$\begin{aligned}
S(p_i, p_j) &= \frac{x^\delta}{\text{LT}(p_i)} p_i - \frac{x^\delta}{\text{LT}(p_j)} p_j \\
&= \frac{x^\delta}{c_i x^{\alpha_i} \text{LT}(g_i)} c_i x^{\alpha_i} g_i - \frac{x^\delta}{c_j x^{\alpha_j} \text{LT}(g_j)} c_j x^{\alpha_j} g_j \\
&= \frac{x^\delta}{\text{LT}(g_i)} g_i - \frac{x^\delta}{\text{LT}(g_j)} g_j \\
&= x^{\delta-\gamma_{ij}} \underbrace{\left(\frac{x^{\gamma_{ij}}}{\text{LT}(g_i)} g_i - \frac{x^{\gamma_{ij}}}{\text{LT}(g_j)} g_j \right)}_{S(g_i, g_j)},
\end{aligned}$$

where $\gamma_{ij} = \text{lcm}(\text{LM}(g_i), \text{LM}(g_j))$, we can write it as a linear combination of $x^{\delta-\gamma_{ij}} S(g_i, g_j)$. By the definition of S-polynomials, $S(g_i, g_j)$ is divisible by G , we have

$$S(g_i, g_j) = \sum_{k=1}^t A_k g_k,$$

where $A_k \in \mathbb{F}[x_1, \dots, x_n]$ and $\text{multideg}(A_k g_k) \leq \text{multideg}(S(g_i, g_j))$. By multiplying $x^{\delta-\gamma_{ij}}$, we have

$$x^{\delta-\gamma_{ij}} S(g_i, g_j) = \sum_{k=1}^t B_k g_k,$$

where $B_k = x^{\delta-\gamma_{ij}} A_k$ and $\text{multideg}(B_k g_k) \leq \text{multideg}(x^{\delta-\gamma_{ij}} S(g_i, g_j)) < \delta$. Hence we can write the first term of (2.1)

$$\sum_{\text{multideg}(h_i g_i) = \delta} \text{LT}(h_i) g_i = \sum_{k=1}^t \tilde{B}_k g_k,$$

where $\tilde{B} = \sum_{k=1}^t B_k$ and $\text{multideg}(\tilde{B}_k g_k) < \delta$. Altogether, we write f as

$$\sum_{\text{multideg}(h_i g_i) = \delta} \text{LT}(h_i) g_i + \sum_{\text{multideg}(h_i g_i) = \delta} (h_i - \text{LT}(h_i)) g_i + \sum_{\text{multideg}(h_i g_i) < \delta} h_i g_i,$$

which is a polynomial combination of g_i but all terms have multidegree less than δ . This contradicts the assumption of δ is minimal, therefore the multidegree of f must be δ . □

Example 2.15

Consider the ideal $I = \langle x - z^2, y - z^3 \rangle$ and basis $G = \{x - z^2, y - z^3\}$ with lex order.

$$\begin{aligned}
S(x - z^2, y - z^3) &= \frac{xy}{x}(x - z^2) - \frac{xy}{y}(y - z^3) \\
&= xz^3 - yz^2 \\
&= z^3(x - z^2) - z^2(y - z^3).
\end{aligned}$$

By Buchberger's Criterion, G is a Gröbner basis.

2.6 Buchberger's Algorithm

The Hilbert Basis Theorem tells us every ideal has a Gröbner basis, but how do we actually construct a Gröbner basis? To see the main idea behind the method, we first revisit Example 2.6. Consider the basis $F = \{f_1, f_2\}$ with grlex order, where $f_1 = x^3 - 2xy$ and $f_2 = x^2y - 2y^2 + x$, and let $I = \langle f_1, f_2 \rangle$. We have seen previously that $\langle \text{LT}(I) \rangle \neq \langle \text{LT}(f_1), \text{LT}(f_2) \rangle$, thereby F is not a Gröbner basis. Alternatively,

$$S(f_1, f_2) = \frac{x^3y}{x^3}(x^3 - 2xy) - \frac{x^3y}{x^2y}(x^2y - 2y^2 + x) = -x^2$$

is not divisible by F , by Buchberger's Criterion, F is not a Gröbner basis. A natural idea is to extend F such that the S-polynomials are divisible by F :

1. Set $F = \{f_1, f_2\}$, we have

- (1) $S(f_1, f_2) = -x^2$ is not divisible by F .

Set $f_3 = -x^2$.

2. Set $F = \{f_1, f_2, f_3\}$, we have

- (1) $S(f_1, f_2)$ is divisible by F ,
 - (2) $S(f_1, f_3) = -2xy$ is not divisible by F ,
 - (3) $S(f_2, f_3) = -2y^2 + x$ is not divisible by F .

Set $f_4 = -2xy$.

3. Set $F = \{f_1, f_2, f_3, f_4\}$, we have

- (1) $S(f_1, f_2)$ and $S(f_1, f_3)$ are divisible by F ,
 - (2) $S(f_1, f_4) = -2xy^2$ is divisible by F ,
 - (3) $S(f_2, f_3) = -2y^2 + x$ is not divisible by F ,
 - (4) one can also check $S(f_2, f_4)$ or $S(f_3, f_4)$.

Set $f_5 = -2y^2 + x$.

4. Set $F = \{f_1, f_2, f_3, f_4, f_5\}$, one can check that $S(f_i, f_j)$ for all $1 \leq i < j \leq 5$ are divisible by F , by Buchberger's Criterion, F is a Gröbner basis.

The example suggests that if any S-polynomial has a nonzero remainder, we add it to F . The full algorithm is depicted in Algorithm 2. However, the Gröbner basis computed by Buchberger's Algorithm is often bigger than necessary. We can often eliminate redundant generators.

Lemma 2.16

Let G be a Gröbner basis of $I \in \mathbb{F}[x_1, \dots, x_n]$, and $p \in G$ be a polynomial such that $\text{LT}(p) \in \langle \text{LT}(G \setminus \{p\}) \rangle$. Then $G \setminus \{p\}$ is also a Gröbner basis of I .

Proof. By definition, $\langle \text{LT}(G) \rangle = \langle \text{LT}(I) \rangle$. If $\text{LT}(p) \in \langle \text{LT}(G \setminus \{p\}) \rangle$, then $\langle \text{LT}(G \setminus \{p\}) \rangle = \langle \text{LT}(G) \rangle = \langle \text{LT}(I) \rangle$, thereby $\langle \text{LT}(G \setminus \{p\}) \rangle$ is also a Gröbner basis. $\square$

Algorithm 2 Buchberger's Algorithm

Input: $F = \{f_1, \dots, f_s\}$ **Output:** Gröbner basis $G = \{g_1, \dots, g_t\}$

```
1: initialize  $G = F$ 
2: while true do
3:   set  $G' = G$ 
4:   for  $1 \leq i < j \leq |G'|$  do
5:      $r \leftarrow$  the remainder of the division of  $S(g'_i, g'_j)$  by  $G'$ 
6:     if  $r \neq 0$  then
7:        $G \leftarrow G \cup \{r\}$  and break
8:     end if
9:   end for
10:  if  $G = G'$  then
11:    break
12:  end if
13: end while
14: return  $G$ 
```

By making all leading coefficients to 1 and removing all p such that $\text{LT}(p) \in \langle \text{LT}(G \setminus \{p\}) \rangle$, we arrive at what we call a minimal Gröbner basis. Consider the ideal $I = \langle x^3 - 2xy, x^2y - 2y^2 + x \rangle$, and the Gröbner basis $G = \{f_1, f_2, f_3, f_4, f_5\}$ from the previous example. We first adjust the coefficients

$$\begin{array}{ll} f_1 = x^3 - 2xy & \tilde{f}_1 = x^3 - 2xy \\ f_2 = x^2y - 2y^2 + x & \tilde{f}_2 = x^2y - 2y^2 + x \\ f_3 = -x^2 & \implies \tilde{f}_3 = x^2 \\ f_4 = -2xy & \tilde{f}_4 = xy \\ f_5 = -2y^2 + x & \tilde{f}_5 = y^2 - (1/2)x \end{array}$$

Since $\text{LT}(f_1) = x^3 = x\text{LT}(\tilde{f}_3)$ and $\text{LT}(f_2) = x^2y = x\text{LT}(\tilde{f}_4)$, by Lemma 2.16, the set $\{\tilde{f}_3, \tilde{f}_4, \tilde{f}_5\}$ is a minimal Gröbner basis of I . When G is a minimal Gröbner basis, by Proposition 2.5, the leading terms $\{\text{LT}(p) \mid p \in G\}$ forms a minimal basis of $\langle \text{LT}(I) \rangle$. However, the ideal I can have many minimal Gröbner bases. We next define a particular minimal Gröbner basis.

Definition 2.17: Reduced Gröbner Basis

A reduced Gröbner basis for a polynomial ideal I is a Gröbner basis G of I such that for all $p \in G$, the leading coefficient $\text{LC}(p) = 1$ and no monomial of p lies in $\langle \text{LT}(G \setminus \{p\}) \rangle$.

A nice property is that let $I \neq \{0\}$ be a polynomial ideal, I has a unique reduced Gröbner basis. We now answer the solutions to the problems at the start of this section:

1. Problem 1: Every ideal has a Gröbner basis.
2. Problem 2: Find a Gröbner basis G of $I = \langle f_1, \dots, f_s \rangle$, then $f \in I$ if and only if f is divisible by G .
3. Problem 3: Given a polynomial system $f_1 = \dots = f_s = 0$, let $I = \langle f_1, \dots, f_s \rangle$, the solution space is $V(f_1, \dots, f_s) = V(I)$. We find a Gröbner basis $G = \{g_1, \dots, g_t\}$, and since $V(f_1, \dots, f_s) = V(g_1, \dots, g_t)$, we solve the polynomial system $g_1 = \dots = g_t = 0$.
4. Problem 4: First write the parametric representation into a polynomial equation system, then use Gröbner basis and eliminate the parameters.

References

- [1] D. Cox, J. Little, and D. O'shea. *Ideals, varieties, and algorithms*. Springer, 1997.